



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES



CAPSTONE PROJECT PRESENTATION
CSA1707 - ARTIFICIAL INTELLIGENCE

AI-Powered Electrical Panel Fault Prediction
and Preventive Maintenance System

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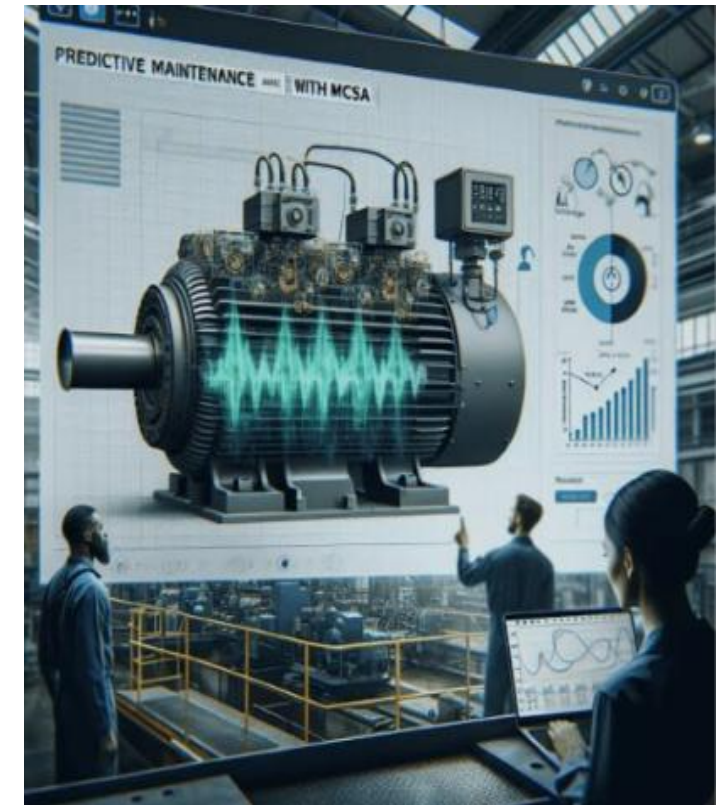


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ABSTRACT



- The system monitors the temperature and current of an electrical panel continuously using sensors.
- An ESP32 collects and processes the real-time sensor data.
- It identifies overheating and abnormal current conditions that may indicate potential faults.
- The system analyzes abnormal patterns to provide early fault prediction and warning.
- The proposed solution supports predictive maintenance, improved safety, and Industry 4.0 applications



INTRODUCTION



- Electrical panels are essential for safe and reliable power distribution in industries, buildings and commercial facilities.
- Common problems include overheating, overload, abnormal voltage/current, loose connections and component degradation.
- Traditional maintenance is mainly scheduled or reactive and may not identify developing faults early.
- Artificial Intelligence can learn patterns from sensor data and predict abnormal conditions before a major failure.

PROBLEM STATEMENT

- Electrical panel faults such as overheating and abnormal current can cause equipment damage and unexpected failures.
- Traditional inspection methods may not detect these problems before they become critical.
- Continuous manual monitoring is time-consuming and inefficient for industrial environments.
- There is a need for a low-cost real-time system that can monitor electrical conditions continuously.
- The proposed system aims to detect abnormal conditions early and predict potential faults using ESP32 and sensor-based monitoring.

OBJECTIVES

- To continuously monitor temperature and current in an electrical panel using sensors.
- To use ESP32 for real-time collection and processing of sensor data.
- To detect overheating and abnormal current conditions at an early stage.
- To analyze sensor readings to identify and predict potential electrical faults.
- To improve safety, reliability, and predictive maintenance in Industry 4.0 environments.





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LITERATURE REVIEW



Year	Research / Study	Method / Focus	Key Finding	Limitation
2022	Machine Learning for Predictive Maintenance	ML-based condition monitoring	ML can identify abnormal equipment behavior	Requires sufficient quality data
2023	AI-Based Fault Diagnosis in Electrical Systems	Machine learning & fault classification	AI improves early detection of electrical faults	Model accuracy depends on sensor data
2024	IoT-Based Predictive Maintenance	IoT sensors + real-time monitoring	Continuous monitoring enables early fault identification	Connectivity and hardware cost
2024	Deep Learning for Equipment Fault Prediction	Deep learning & time-series analysis	Learns complex patterns from historical sensor data	Requires larger datasets and computation
2025	AI-Driven Predictive Maintenance Review	AI/ML + condition monitoring	Predictive maintenance can reduce failures and improve maintenance planning	Real-world implementation remains challenging

EXISTING SYSTEM



- Electrical panels are mainly monitored through manual inspection and scheduled maintenance.
- Faults such as overheating, overload, and abnormal current may be detected only after noticeable symptoms occur.
- Traditional systems generally use fixed maintenance schedules instead of predicting the actual condition of equipment.
- Continuous manual monitoring is time-consuming and may miss developing faults.
- These limitations can lead to unexpected failures, equipment damage, downtime, and higher maintenance costs.

PROPOSED SYSTEM

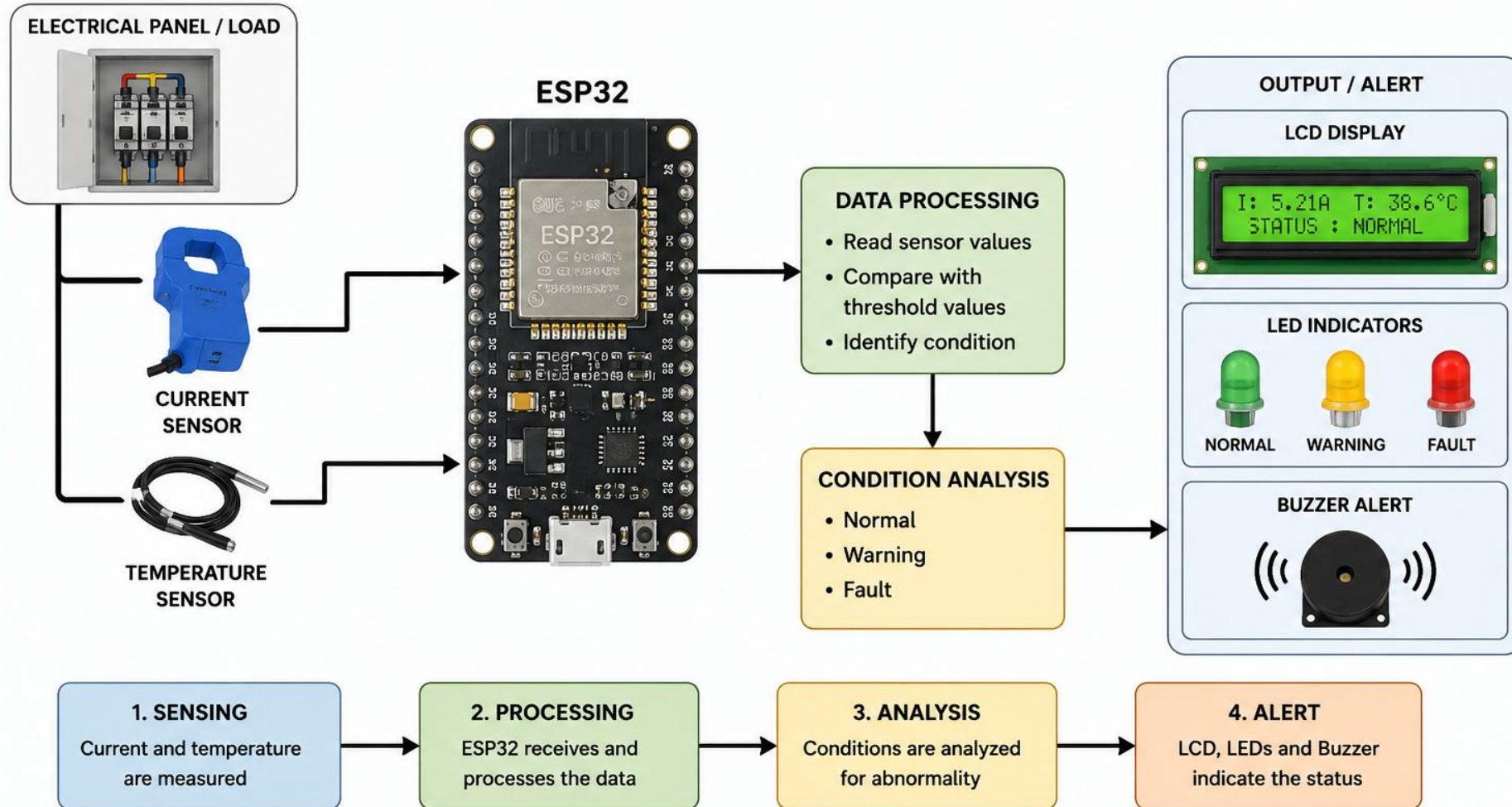


- The proposed system uses an ESP32-based monitoring unit to continuously monitor electrical panel conditions.
- A current sensor measures the electrical current, while a temperature sensor monitors the panel temperature.
- The ESP32 collects and processes the sensor readings in real time.
- The system analyzes the readings to identify normal, warning, and abnormal/fault conditions.
- The system helps enable early fault detection, preventive maintenance, improved safety, and reduced downtime.



WORKING PRINCIPLE

The system continuously monitors current and temperature of the electrical panel, processes the data and provides alert when abnormal conditions are detected.



MODULE 1 :SENSING & DATA COLLECTION

Objective :

To collect real-time electrical and thermal data from the electrical panel using sensors.

Content:

- The current sensor measures the current flowing through the electrical load.
- The temperature sensor continuously monitors the temperature of the electrical panel.
- Both sensors provide real-time readings to the ESP32.
- The ESP32 collects and processes the sensor data for further analysis.
- The collected data acts as the primary input for detecting abnormal conditions and predicting potential faults.

MODULE 2: AI FAULT PREDICTION

Objective:

To process the sensor readings and identify abnormal electrical and thermal conditions.

Content:

- The ESP32 receives the current and temperature readings from the sensors.
- The sensor values are processed and continuously monitored in real time.
- The system compares the readings with the defined normal operating conditions.
- Abnormal changes in current or temperature are identified as potential fault conditions.
- The system classifies the panel condition as Normal, Warning, or Fault.
- The processed data supports early prediction of possible electrical failures.

MODULE 3: PREVENTIVE MAINTENANCE

Objective:

To display the system condition and provide an alert when an abnormal condition or potential fault is detected.

Content:

- The processed sensor values and system status are displayed on the LCD.
- LED indicators provide a visual indication of the panel condition.
- A buzzer provides an audible alert when a fault condition is detected.
- The system indicates different conditions such as Normal, Warning, and Fault.
- Early alerts help the user take preventive action before a major electrical failure occurs.

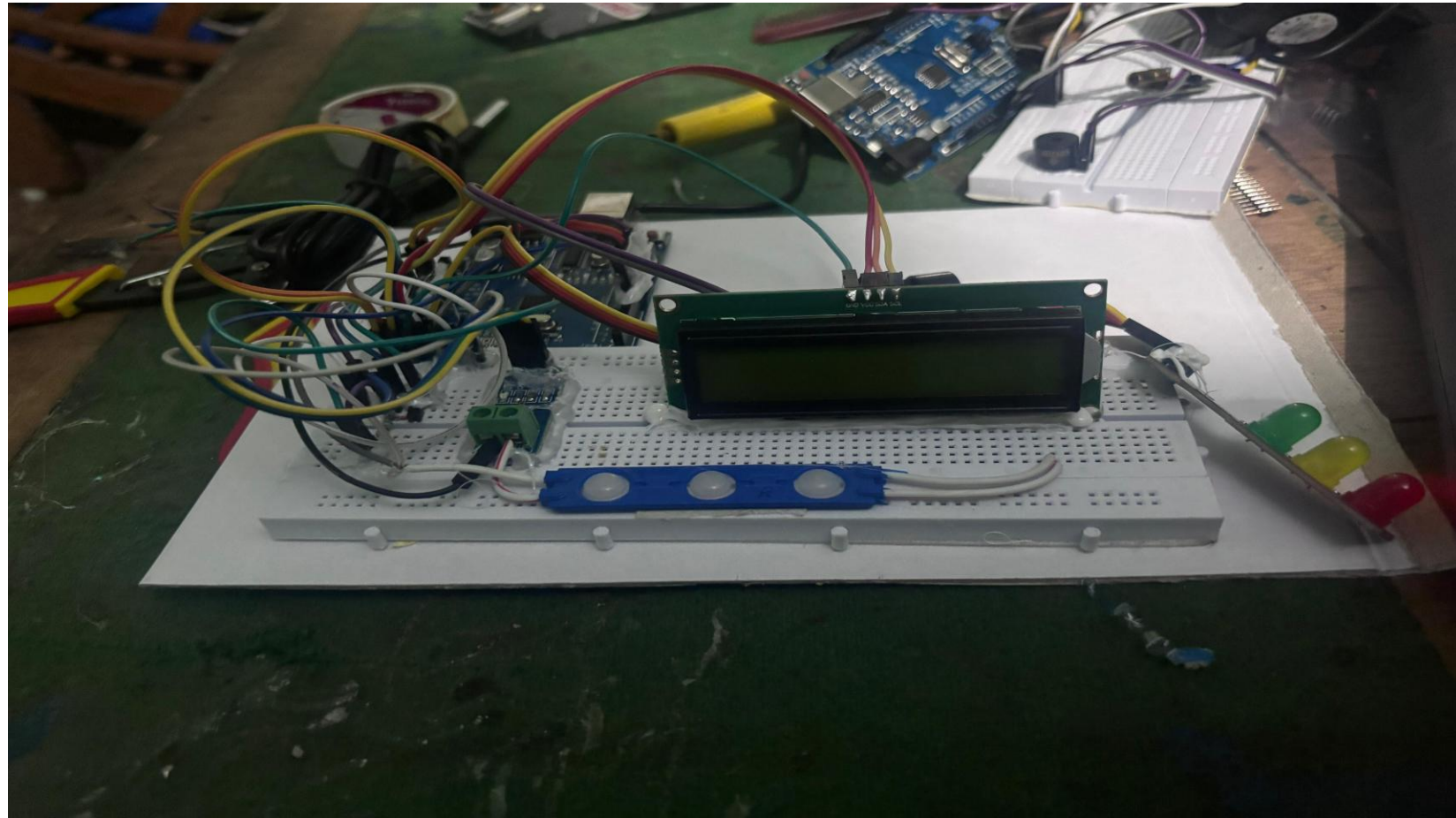
DATASET

- **Dataset:** Real-time sensor data collected from the electrical panel.
- **Parameters:** The dataset contains current (A), temperature ($^{\circ}\text{C}$), timestamp, and fault condition.
- **Data Collection:** An ESP32 collects current and temperature readings continuously.
- **Algorithm Used:** Threshold-based rule classification is used to identify Normal, Warning, and Fault conditions.
- **Future AI/ML:** An AI/ML model can be integrated later using historical sensor data for more accurate fault prediction.

METHODOLOGY

- Sensor Data Collection – The current and temperature sensors continuously collect real-time data from the electrical panel.
- ESP32 Data Acquisition – The ESP32 receives the sensor readings and processes the collected data.
- Data Analysis – The current and temperature values are monitored to identify abnormal operating conditions.
- Fault Prediction – Abnormal patterns such as excessive current or overheating are identified as potential fault conditions.
- Alert & Indication – The system displays the status on the LCD and provides LED/buzzer alerts when a warning or fault is detected.

IMPLEMENTATION



RESULTS & DISCUSSION

Real-Time Monitoring

Continuous monitoring of current and temperature

Condition Detection

Identifies normal and abnormal electrical conditions

Fault Alerting

LED, LCD and buzzer indicate warning or fault status

Preventive Action

Early alerts help prevent equipment damage and unexpected failure

FUTURE SCOPE

- Integrate an AI/ML model for more accurate fault prediction.
- Add Wi-Fi and cloud connectivity for remote monitoring.
- Develop a web/mobile dashboard with real-time notifications.
- Store sensor data for historical analysis and fault trends.
- Expand monitoring to include voltage, power, vibration, and multiple electrical panels.



CONCLUSION

- The proposed system successfully provides **real-time monitoring of electrical panel current and temperature**.
- The **ESP32** effectively collects and processes sensor readings to identify abnormal conditions.
- The system detects conditions such as **overheating and abnormal current** at an early stage.
- **LCD, LEDs, and buzzer alerts** provide clear indication of the panel status.
- The system supports **predictive maintenance, improved safety, and reduced risk of unexpected electrical failures** in Industry 4.0 environments.

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THANK YOU